Week 10 Notes

STAT 251 SECTION 01

From estimating to testing

For the last two weeks we have been **estimating** parameters: a confidence interval answers the question *"what is p (or μ), give or take?"* A lot of questions are not asked that way. They come as a **claim**, and we want to know whether the data back it up:

- A new drug claims to lower blood pressure. Does it, or did the patients' readings just happen to drop?
- The polling averages say a Senate candidate has 52% support. A new poll finds less. Has support really dropped, or is the new poll just a different sample?
- Your roommate says they can tell Coke from Pepsi. Can they, or are they guessing?

Each claim is about a **parameter**: the drug's average effect μ, the candidate's support p, your roommate's chance p of naming a cup correctly. We never see the parameter. We see an **estimate** from a sample, and estimates bounce around from sample to sample (Week 8: two polls of the same state give two different answers). So the real question is always:

> **Is the difference between what we saw and what the claim says too big to be ordinary sampling variation?**

A **significance test** (also called a **hypothesis test**) is the tool that answers it. A result that is decidedly **not** due to ordinary variation, that is, not due to chance or coincidence, is called **statistically significant**.

The logic of a significance test

In plain words: a courtroom

A significance test works like a criminal trial.

- The defendant is **presumed innocent**. Nobody has to prove innocence; innocence is the starting assumption. In a test, the starting assumption is the boring explanation, "nothing

interesting is going on", and it is called the **null hypothesis**.

- The prosecution has a **suspicion**: the defendant did it. In a test, the suspicion is called the **alternative hypothesis**. The burden of proof is on the suspicion, not on the boring explanation.
- The jury asks: **if the defendant really were innocent, how likely is evidence like this?** If that is very unlikely (beyond reasonable doubt), they convict. In a test, that probability is the **P-value**, and "beyond reasonable doubt" is a cutoff we pick in advance, the **significance level** α.
- A jury that does not convict says **"not guilty"**, never "innocent". Weak evidence does not prove innocence. In a test, we either **reject** the null hypothesis or **do not reject** it. We never "accept" it.

Hold on to that picture. Everything below is the same four ideas with numbers attached.

Working it through: can you taste the difference?

Your roommate claims they can tell Coke from Pepsi. You set up a blind test: 8 times, you flip a coin to decide which soda to pour, and they taste the cup and name it. They get **all 8 right**. Impressive, or lucky?

Step 1: name the parameter, and the two stories

The parameter is p, your roommate's true, long-run chance of naming a cup correctly. We cannot see p; we only see this one test, where they got $\hat{p} = 8/8 = 1$. There are two stories about p:

- **Story 1, "just guessing":** they cannot really tell, so each cup is a coin flip, and $p = 0.5$. This is the **null hypothesis**, written $H_0 : p = 0.5$.
- **Story 2, "they can taste it":** they do better than guessing, $p > 0.5$. This is the **alternative hypothesis**, written $H_A : p > 0.5$.

Notice that the hypotheses are about p, the parameter, not about $\hat{p}$. We already **know** $\hat{p} = 1$; there is nothing to test about it. The question is what it tells us about p.

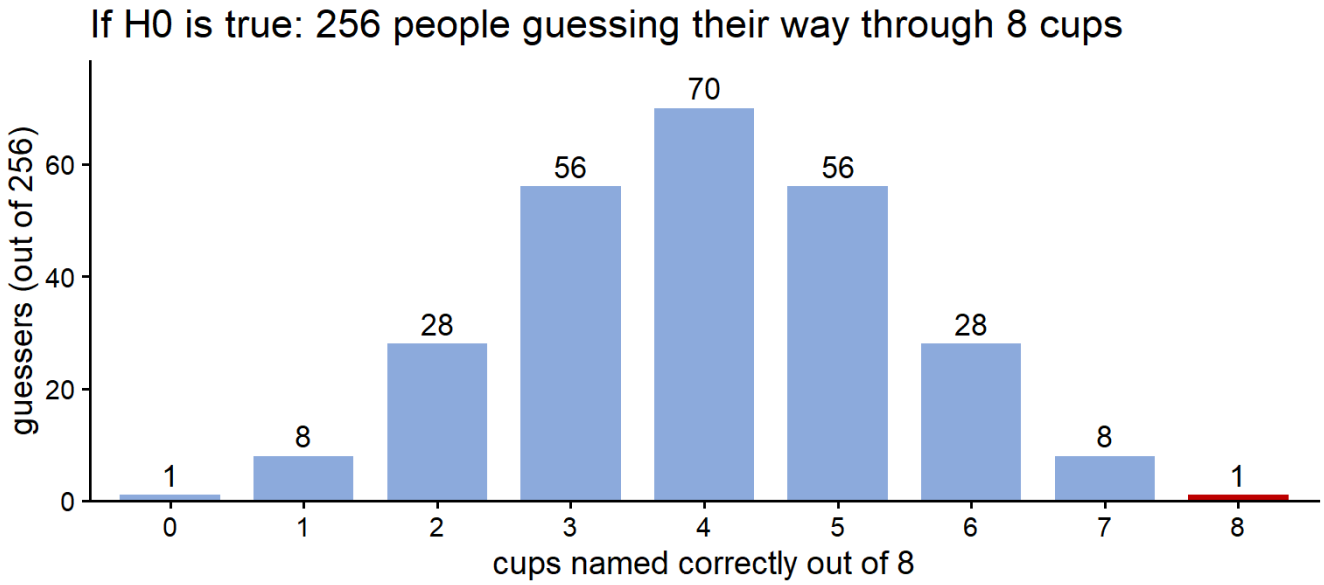
Step 2: suppose H_0 is true. What results would we expect?

Here is the key move. We cannot prove story 2 directly, so we **assume story 1** and work out what it predicts. If they are only guessing, then the number of cups they get right, X, is a count of successes in 8 independent trials that each succeed with probability 0.5. That is a binomial

random variable from Week 7:

$$X \sim \text{Binom}(8,\ 0.5) \qquad \text{when } H_0 \text{ is true}$$

A helpful way to see this distribution: imagine **256 people who cannot taste the difference** each take the 8-cup test, guessing every time. Since $2^8 = 256$, the binomial formula $P(X = k) = \binom{8}{k}(0.5)^k(0.5)^{8-k} = \binom{8}{k}/256$ says that, on average, exactly $\binom{8}{k}$ of them score k:



This is what "just guessing" looks like. Most guessers land near 4 out of 8, which makes sense: the mean of a binomial is $np = 8(0.5) = 4$. Some get 5 or 6 by luck. Getting **all 8** happens to about **1 guesser in 256**.

Step 3: how surprising is what we actually saw?

Now compare the data with the prediction. The probability, **if H_0 is true**, of a result at least as extreme as the one we saw is called the **P-value**. "At least as extreme" means at least as far in the direction H_A points to, which here is "as many correct, or more":

$$\text{P-value} = P(X \geq 8 \mid H_0 \text{ true}) = \binom{8}{8}(0.5)^8 = \frac{1}{256} \approx 0.0039$$

The bar $\mid$ is read **"given"**: the probability is worked out in the world where H_0 is true. That is the same conditional probability idea from Week 6.

A P-value of 0.0039 says: **if your roommate were only guessing, a result this good would happen less than half a percent of the time.**

Step 4: two explanations, and a decision

There are only two ways to explain a result with a small P-value:

1. **Something unusual happened.** Your roommate was guessing and got very lucky: they are the 1 in 256.
2. **The assumption behind the calculation is wrong.** H_0 is false; they really can taste the difference.

Both are possible. The test does not tell us which one is true. It tells us how hard we would have to believe in luck to keep believing H_0. The smaller the P-value, the harder that gets, and at some point explanation 2 becomes the more reasonable one.

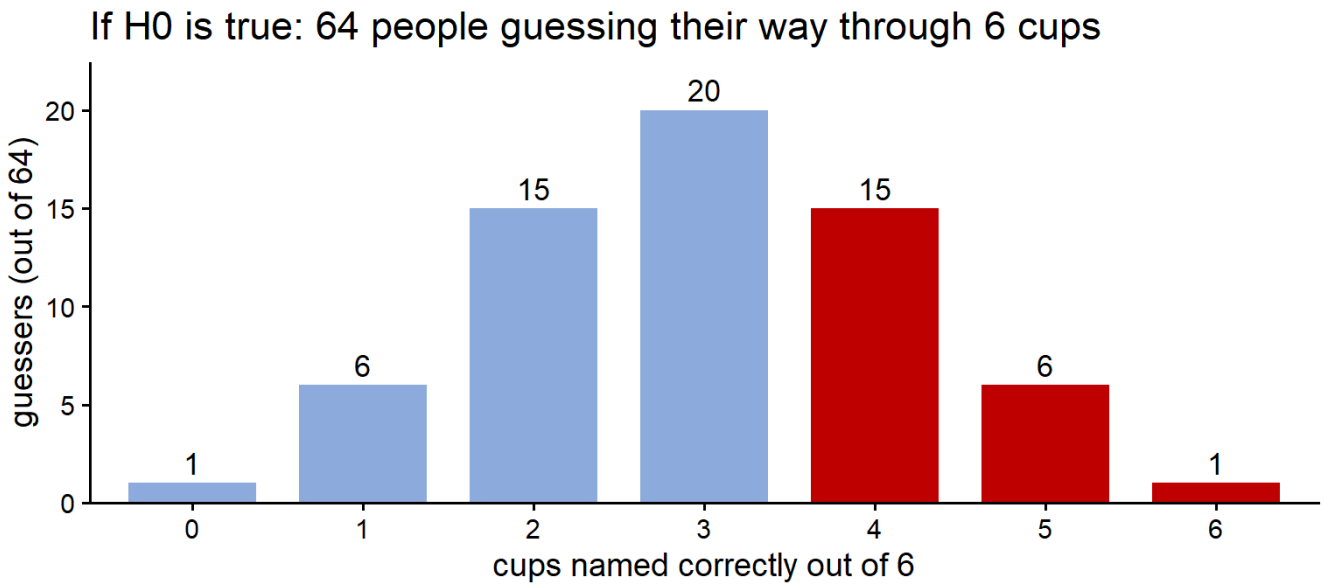
Where is that point? It is a choice, made **before** looking at the data, called the **significance level** α. By convention $\alpha = 0.05$, i.e. "if a result this extreme would happen less than 5% of the time by luck alone, we stop believing it was luck". The decision rule is:

- If the P-value $< \alpha$: **reject** H_0. The result is statistically significant; the data are evidence for H_A.
- If the P-value $\geq \alpha$: **do not reject** H_0. The result is not statistically significant; the data are not strong enough evidence for H_A.

Here $0.0039 < 0.05$, so we reject H_0: the data are evidence that your roommate can taste the difference. The textbook's memory device: **"if the P-value is low, the null must go."**

A second taster

A friend makes the same claim, but takes a shorter 6-cup test and gets **4 of 6** right. Under $H_0 : p = 0.5$, their score is $X \sim \text{Binom}(6,\ 0.5)$. Picture 64 guessers ($2^6 = 64$):



"At least as extreme" as 4 correct means 4, 5 or 6 correct, the shaded bars:

$$\text{P-value} = P(X \geq 4 \mid H_0 \text{ true}) = \frac{15+6+1}{64} = \frac{22}{64} \approx 0.34$$

About a third of pure guessers do this well, so 4 of 6 is not surprising at all if your friend is guessing. $0.34 \geq 0.05$: **do not reject** H_0. This does **not** prove your friend is guessing ("not guilty", not "innocent"). It says this test gives no real evidence that they can taste the difference. A longer test might.

One more thought. If all 256 guessers from the first picture took the 8-cup test, about one of them would score 8 for 8 and be convinced they have a gift. A small P-value means **either** rare luck **or** a false H_0, and from one test we cannot be sure which. That is why surprising results get repeated before anyone believes them.

The notation

Significance tests bring a lot of new symbols. Most of them are old ideas with new names.

Symbol	Read as	Meaning
θ	"theta"	a parameter, any parameter: stands for p or μ when we talk about tests in general
θ_0, e.g. p_0, μ_0	"theta-naught", "p-naught", "mu-naught"	the value of the parameter that H_0 claims, a number like 0.5 or 28 g

Symbol	Read as	Meaning
$\hat{\theta}$, e.g. $\hat{p}$, $\bar{x}$	"theta-hat", "p-hat", "x-bar"	the estimate of θ from our sample (Weeks 1 and 8)
$H_0 : \theta = \theta_0$	"H-naught: theta equals theta-naught"	the null hypothesis: "no effect", the parameter equals the claimed value
$H_A : \theta > \theta_0$	"H-A: theta is greater than theta-naught"	an upper-tailed alternative: we suspect the parameter is bigger
$H_A : \theta < \theta_0$	"H-A: theta is less than theta-naught"	a lower-tailed alternative: we suspect the parameter is smaller
$H_A : \theta \neq \theta_0$	"H-A: theta is not equal to theta-naught"	a two-sided alternative: we suspect it is different, in either direction
z_{obs}	"z observed"	the test statistic: how many standard errors $\hat{\theta}$ is from θ_0
$P(Z \geq z_{obs} \mid H_0 \text{ true})$	"the probability Z is at least z observed, given H-naught is true"	the P-value of an upper-tailed test
α	"alpha"	the significance level, the cutoff for the P-value, usually 0.05

The subscript 0 always marks "the value from the null hypothesis". In the taste test, θ is p, $\theta_0 = p_0 = 0.5$, and $\hat{\theta} = \hat{p}$.

Writing the hypotheses

Hypotheses come first, before any data are looked at. A good order to write them in:

1. **Start with H_A**: what do we suspect, or hope to find evidence for? "They can taste the difference" becomes $H_A : p > 0.5$.
2. **Then write H_0** as "that effect is not there", with an equals sign: $H_0 : p = 0.5$.

Rules that always hold:

- Both hypotheses are about a **parameter** (p or μ), never an estimate. $H_0 : \hat{p} = 0.5$ makes no sense: we already know $\hat{p}$.
- H_0 always contains the **equals sign**, and H_A never does.

- H_A must come from the question, **not from the data**. Looking at the data first and then pointing H_A the way the data went is a form of bias: it makes the evidence look stronger than it is.

The direction of H_A decides which kind of test it is:

Alternative	Name	Use it when the question says
$H_A : \theta > \theta_0$	upper-tailed (one-sided)	"more than", "better", "increased", "at least"
$H_A : \theta < \theta_0$	lower-tailed (one-sided)	"less than", "worse", "decreased", "fewer"
$H_A : \theta \neq \theta_0$	two-sided	"different", "changed", "is it fair?"

Example: stating hypotheses

For each story, decide what the parameter is, write H_A first (the suspicion), then H_0 (no effect), give θ_0 and say which tail.

a. *The taste test.* Your roommate swears they can tell Coke from Pepsi with their eyes closed. You decide to put it to the test: behind their back, you flip a coin to choose which soda goes in each cup, hand them the cups one at a time, and write down whether they name each one correctly. Someone who is just guessing would be right about half the time. Does your roommate do better than guessing?

Answer: p = their chance of naming a cup right. $H_A : p > 0.5$, $H_0 : p = 0.5$, with $p_0 = 0.5$: **upper-tailed**. Only "better than guessing" would support the claim.

b. *The spinning coin.* A friend says that a penny spun on its edge on a table does not land heads half the time, the way a flipped penny does. Neither of you knows which way it should lean: the two faces are stamped differently, so the coin might fall toward heads or toward tails. You plan to spin it many times and count the heads. Is a spun penny fair?

Answer: p = the chance a spun penny lands heads. We do not suspect a direction, only "not fair": $H_A : p \neq 0.5$, $H_0 : p = 0.5$, with $p_0 = 0.5$: **two-sided**.

c. *The chip bag.* Every bag of your favorite chips says "Net weight 28 g". Lately the bags seem to have more air and fewer chips, and you suspect the company is quietly under-filling them. You buy a bag from several different stores and weigh the chips in each one. Do the bags hold less than 28 g on average?

Answer: μ = the mean weight of chips in all bags of this brand. $H_A : \mu < 28$, $H_0 : \mu = 28$, with $\mu_0 = 28$ g: **lower-tailed**. Only an under-filled bag would cheat the customer.

d. *The coffee machine.* The campus coffee shop's machine is set to pour 12 oz into a medium cup. If it pours too little, customers complain; if it pours too much, the shop wastes coffee and cups overflow. Each month the manager measures a sample of medium pours to check that the machine is still set correctly. Is the machine's mean pour off from 12 oz?

Answer: μ = the mean amount the machine pours into a medium cup. A problem in either direction matters: $H_A : \mu \neq 12$, $H_0 : \mu = 12$, with $\mu_0 = 12$ oz: **two-sided**.

Measuring "how surprising": the test statistic

For the taste test we could list every possible score and add up the binomial probabilities. With hundreds or thousands of observations that gets impractical, but we already have a better tool: the **sampling distribution** and the **central limit theorem** from Week 8.

Recall what the CLT told us about $\hat{p}$: over all possible samples of size n, the values of $\hat{p}$ pile up in a roughly normal shape, centered at the true p, with standard error $SE = \sqrt{p(1-p)/n}$. In a significance test we **assume H_0 is true**, so the true p is p_0. That pins down the whole pile:

$$\text{if } H_0 \text{ is true:} \qquad \hat{p} \;\approx\; N\left(p_0,\ \sqrt{\frac{p_0(1-p_0)}{n}}\right)$$

Notice where every number comes from. The **center** p_0 comes from H_0, not from the data. The standard error also uses p_0, because we are describing the world where H_0 is true. The data enter only once, as the single $\hat{p}$ we observed, which we then place on that curve.

To see how far out on the curve it lands, we turn it into a z-score, exactly as in Weeks 6 and 9 (observation minus mean, divided by standard deviation). This z-score is the **test statistic**:

$$z_{obs} = \frac{\hat{\theta} - \theta_0}{SE} \qquad \text{for a proportion:} \qquad z_{obs} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

It measures **how many standard errors the estimate is from the value H_0 claims**. If H_0 is true, z_{obs} comes from the standard normal $N(0,1)$, so values near 0 are ordinary and values beyond about ± 2 are rare. The P-value is then an area under the standard normal curve, found with the z-table just as in Week 9.

This is the same machinery as a confidence interval, pointed in a different direction:

	Confidence interval	Significance test
Question	What is the parameter, give or take?	Is the claim $\theta = \theta_0$ believable?
Normal curve centered at	the estimate $\hat{\theta}$	the claimed value θ_0
Uses	$\hat{\theta} \pm z \times SE$	$z_{obs} = (\hat{\theta} - \theta_0)/SE$
Result	a range of believable values for θ	a P-value, then reject or do not reject H_0

Example: has the candidate's support dropped?

It is midterm season (Election Day is Tuesday, November 3). Suppose the polling averages put Candidate X in a Senate race at **52%** support among likely voters. Your campaign suspects X's support is actually lower. You run your own poll of a random sample of $n = 1{,}000$ likely voters, and **490** say they will vote for X: $\hat{p} = 0.490$.

49% is lower than 52%. But Week 8 taught us that two polls of the same state give different answers just because they reach different people. Is a 3-point gap **bigger than ordinary poll-to-poll variation**? That is exactly a significance test.

1. **Assumptions.** A random sample of likely voters, and a large enough sample: $np_0 = 1000(0.52) = 520$ and $n(1-p_0) = 480$, both well above 15.
2. **Hypotheses.** p = the proportion of **all** likely voters who support X. We suspect it is lower than the polls say, so $H_A : p < 0.52$ and $H_0 : p = 0.52$, with $p_0 = 0.52$: **lower-tailed**.
3. **Test statistic.** If H_0 is true, polls of 1,000 voters give values of $\hat{p}$ centered at 0.52 with

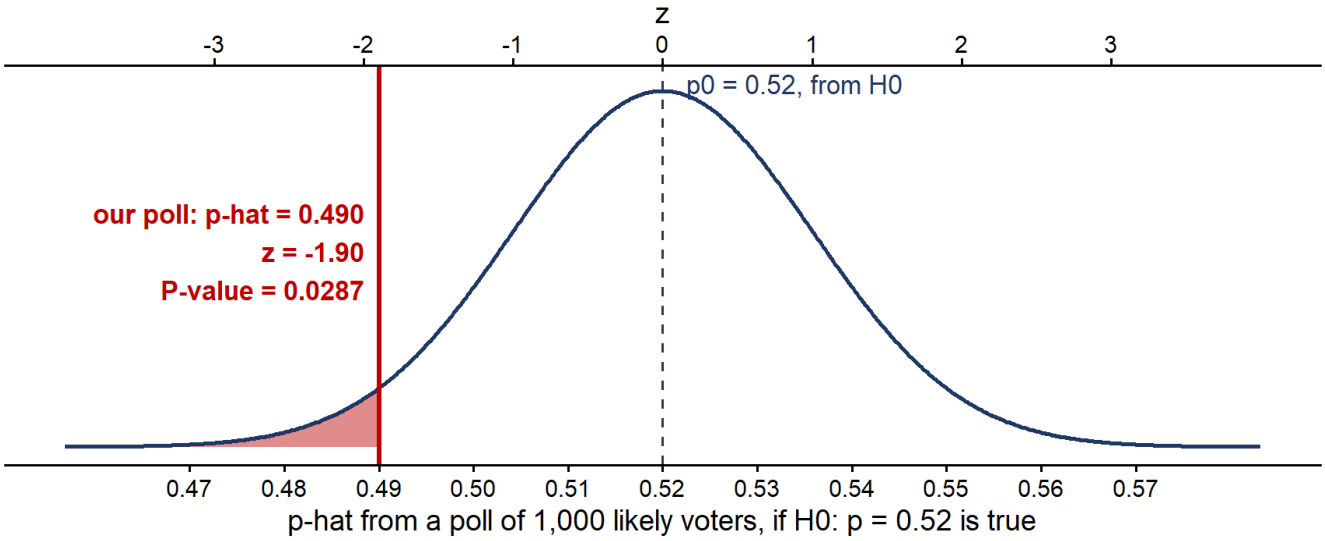
$$SE = \sqrt{\frac{0.52 \times 0.48}{1000}} = \sqrt{0.0002496} \approx 0.0158$$

   so

$$z_{obs} = \frac{0.490 - 0.52}{0.0158} \approx -1.90$$

   Our poll is 1.9 standard errors below what H_0 predicts.
4. **P-value.** $P(Z \le -1.90 \mid H_0 \text{ true}) = 0.0287$, from the z-table. If X's support really were 52%, only about 3 polls in 100 would come out this low.
5. **Decision.** $0.0287 < 0.05$: **reject** H_0. The poll is evidence that X's support is below 52%.

Here is the same reasoning as a picture. The curve shows where $\hat{p}$ from a poll of 1,000 voters would land **if H_0 were true**; the red line is where ours landed, and the shaded area is the P-value.

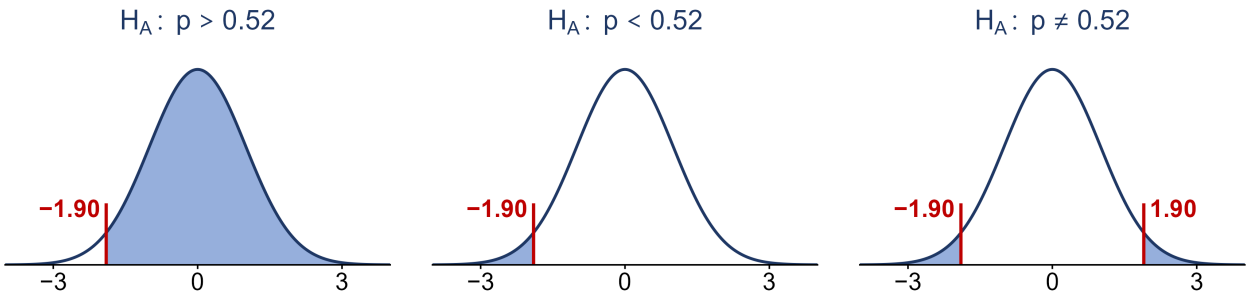


Read the two axes together: the top axis is in the units of the problem (proportions of voters), the bottom axis is the same picture in standard errors. The test statistic is just the poll's position on the bottom axis.

A different claim, the same poll. The headline question on election night is whether X is **losing**, i.e. below 50%. That is a different H_0: $H_0 : p = 0.5$, $H_A : p < 0.5$. Now $SE = \sqrt{0.25/1000} \approx 0.0158$ and $z_{obs} = (0.490 - 0.5)/0.0158 \approx -0.63$, so the P-value is $P(Z \leq -0.63) = 0.2643$. **Do not reject H_0.** The same 49% that is good evidence against "52%" is weak evidence against "50%", because 0.490 sits only 0.63 standard errors below 0.5. A test always answers a question about one specific value θ_0: change θ_0 and you change the question. This is why poll reports say a race is "within the margin of error".

Which tail is the P-value?

"At least as extreme" always means "at least as far in the direction H_A points to". So the shape of H_A decides which area of the curve is the P-value. To see how much it matters, take our poll ($z_{obs} = -1.90$ against $p_0 = 0.52$) and work out the P-value for all three alternatives:



- **Lower-tailed**, $H_A : p < 0.52$: the area **below** z_{obs}, $P(Z \leq -1.90 \mid H_0 \text{ true}) = 0.0287$, read straight from the table.
- **Two-sided**, $H_A : p \neq 0.52$: a poll far out in **either** direction counts as extreme, so the P-value is **both** tails, twice the one-tail area: $2 \times 0.0287 = 0.0574$.
- **Upper-tailed**, $H_A : p > 0.52$: the area **above** z_{obs}, $P(Z \geq -1.90 \mid H_0 \text{ true}) = 1 - 0.0287 = 0.9713$. A poll **below** 52% is no evidence at all that support is **above** 52%, and the huge P-value says so.

At $\alpha = 0.05$, only the lower-tailed test rejects H_0. The two-sided test, with the same data, does not (0.0574): it spreads its doubt over both directions, so it needs a little more evidence. This is why H_A has to come from the question before the poll is run. Choosing the tail after seeing which way the poll went would make any result look more convincing than it is.

Example: do coins land the way they started?

A physics model of coin tossing (Diaconis, Holmes and Montgomery, 2007) predicts that a flipped coin lands on the **same side it started on** a little more than half the time, about 51%, because a real flip wobbles. Bartoš et al. (2025) tested it: 48 people flipped coins 350,757 times, recording the side facing up before and after every flip (and filming it).

Who flipped	Flips
students, for a thesis project	75,036
coin-flipping marathons (up to 12 hours)	203,440
volunteers recruited online	72,281
total	**350,757**

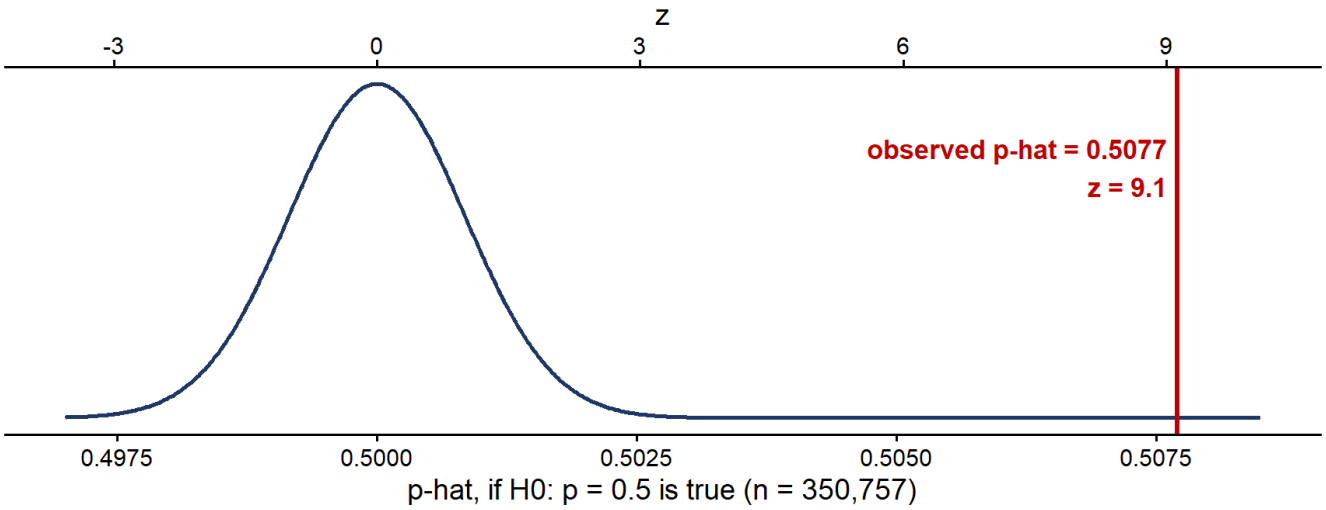
X = 1 if a flip lands on the side it started on, and 0 otherwise. In all, **178,079** of the $n = 350{,}757$ flips did. Test the model's prediction at $\alpha = 0.05$, using the five steps.

1. **Assumptions.** The flips are a random sample of how people flip coins, and n is large: np_0 and $n(1-p_0)$ are both about 175,000, far more than enough for the normal curve to apply.
2. **Hypotheses.** p = the chance a flip lands on its starting side. The model predicted the direction before any flips were made, so $H_A : p > 0.5$ and $H_0 : p = 0.5$, with $p_0 = 0.5$: **upper-tailed**.
3. **Test statistic.** $\hat{p} = 178{,}079/350{,}757 \approx 0.5077$ and $SE = \sqrt{0.5 \times 0.5/350{,}757} \approx 0.000844$, so

$$z_{obs} = \frac{0.5077 - 0.5}{0.000844} \approx 9.1$$

   The estimate sits about **9 standard errors** above p_0.
4. **P-value.** $P(Z \geq 9.1 \mid H_0 \text{ true}) \approx 0$. The z-table stops at 3.49, and 9.1 is far beyond it.
5. **Decision.** The P-value is far below 0.05: **reject** H_0. The data are strong evidence that flips land on their starting side more than half the time, just as the model predicted.

Here is what step 3 looks like on the sampling distribution. The curve is where $\hat{p}$ would land if H_0 were true; the red line is where it actually landed.

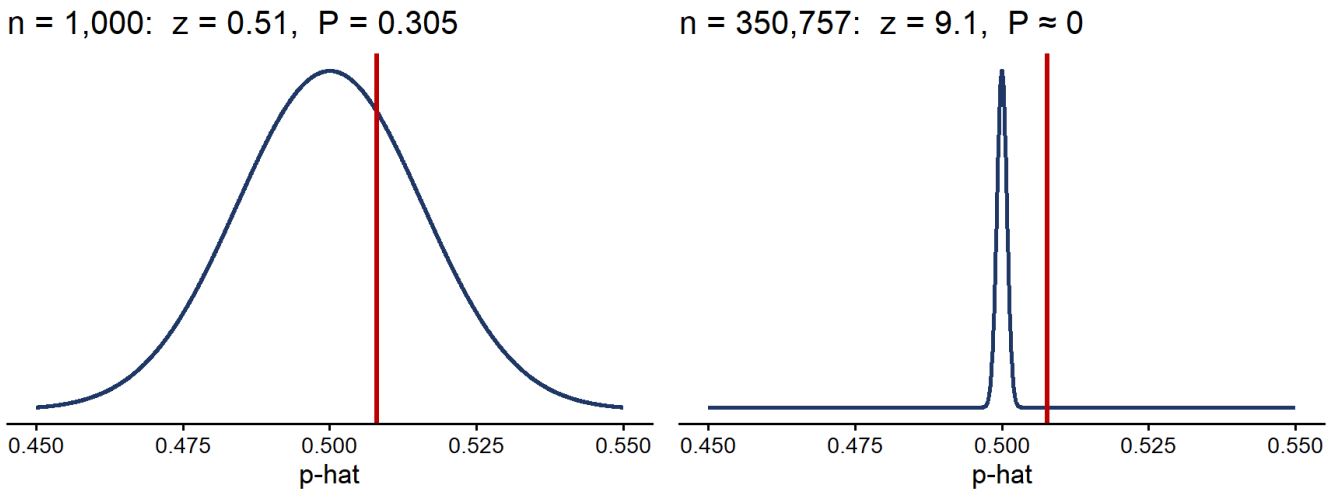


Significant is not the same as important

A 50.8% same-side rate instead of 50% is about **1 extra same-side landing in every 130 flips**. Nobody could notice that in a coin toss before a football game. So how can it be "significant"?

Because significance depends on the **sample size**. The standard error has $\sqrt{n}$ on the bottom, so with 350,757 flips the curve of believable $\hat{p}$ values is extremely narrow, and even a tiny difference lands far out in the tail. Suppose instead only 1,000 flips had been made, with the

same 50.8% (508 same-side). Then $SE = \sqrt{0.25/1000} \approx 0.0158$, $z_{obs} = (0.508 - 0.5)/0.0158 \approx 0.51$, and the P-value is $1 - 0.6950 = 0.3050$: **not** significant.



Same estimate, opposite conclusions. "Statistically significant" means **"probably not chance"**. It does not mean "big" or "important". Whether a difference matters is a question about the context, and a confidence interval (which shows **how big** the effect is) answers it better than a P-value does.

The five steps of a significance test

Every test in the rest of the course follows the same five steps:

1. **Assumptions.** The data come from a random sample or a randomized experiment, plus any conditions on sample size or on the shape of the population.
2. **Hypotheses.** $H_0 : \theta = \theta_0$ and H_A (upper-tailed, lower-tailed or two-sided), written before looking at the data.
3. **Test statistic.** $z_{obs} = (\hat{\theta} - \theta_0)/SE$, computed as if H_0 were true: how many standard errors the estimate is from θ_0.
4. **P-value.** The probability, if H_0 is true, of a test statistic at least as extreme as z_{obs}, in the tail or tails that H_A points to.
5. **Decision.** If the P-value $< \alpha$, reject H_0 (statistically significant); otherwise do not reject H_0. Then say what it means in the context of the question.

"Do not reject H_0" is not the same as "accept H_0". Not rejecting H_0 does not prove it true; the data just are not strong enough evidence against it.

Practice: P-values and decisions

For each test, write the P-value as a probability given H_0 is true. Then use its value to make a decision.

Test	Hypotheses	Test statistic	α	P-value
(a)	$H_0 : p = 0.40$, $H_A : p > 0.40$	$z_{obs} = 1.52$	0.05	0.0643
(b)	$H_0 : \mu = 12$, $H_A : \mu < 12$	$t_{obs} = -2.30$, $df = 15$	0.01	0.0181
(c)	$H_0 : \mu = 50$, $H_A : \mu \neq 50$	$z_{obs} = -2.10$	0.05	0.0358

Answers:

- a. $P(Z \geq 1.52 \mid H_0 \text{ true}) = 0.0643 \geq 0.05$: do not reject H_0.
- b. $P(T \leq -2.30 \mid H_0 \text{ true}) = 0.0181 \geq 0.01$: do not reject H_0. (It would be significant at $\alpha = 0.05$.)
- c. $P(Z \leq -2.10 \text{ or } Z \geq 2.10 \mid H_0 \text{ true}) = 2(0.0179) = 0.0358 < 0.05$: reject H_0.

The same five steps test claims about a population proportion p and about a population mean μ.

Testing a population proportion

When the variable is categorical and the claim is about the proportion of successes, $H_0 : p = p_0$, the five steps become:

1. **Assumptions.** A random sample, and a large enough sample: $np_0 \geq 15$ and $n(1 - p_0) \geq 15$.
2. **Hypotheses.** $H_0 : p = p_0$ against $H_A : p > p_0$, $p < p_0$ or $p \neq p_0$.
3. **Test statistic.**

$$z_{obs} = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

   The standard error uses p_0, not $\hat{p}$, because the whole calculation assumes H_0 is true. (A confidence interval uses $\hat{p}$, because there is no p_0 to assume.)

4. **P-value.** Upper-tailed: $P(Z \geq z_{obs} \mid H_0 \text{ true})$. Lower-tailed: $P(Z \leq z_{obs} \mid H_0 \text{ true})$. Two-sided: double the tail beyond z_{obs}.
5. **Decision.** Compare the P-value with α, then say what it means in context.

The coin-flip study above is one of these tests. Here is a second one, from the textbook.

Example: cell phones and brain cancer

From OpenStax Introductory Statistics 2e, Section 9.5, Example 9.21 (https://openstax.org/books/introductory-statistics-2e/pages/9-5-additional-information-and-full-hypothesis-test-examples).

Almost everyone carries a cell phone, and most of us hold it against our heads every day. That has led to a worry that comes up again and again: can the radio waves from a phone cause brain cancer?

Brain cancer is rare. Among people who do **not** use cell phones, about 0.034% develop it, which is 34 people in every 100,000. If phones raise the risk, then the rate among cell phone users should be higher than 0.034%.

To check, researchers follow a very large group of **420,019 cell phone users**. Over the course of the study, **172** of them develop brain cancer.

If phones made no difference, we would expect about $420{,}019 \times 0.00034 \approx 143$ of these users to develop brain cancer. We saw 172, which is more. But is 172 far enough above 143 to rule out chance? Or is it the kind of number a group this size could produce even if phones are harmless?

That is the question a significance test answers. We use the usual significance level, $\alpha = 0.05$.

Question: At $\alpha = 0.05$, do the data show that cell phone users develop brain cancer at a higher rate than non-users?

1. **Assumptions.** We treat the 420,019 users as a random sample of cell phone users. Each user either develops brain cancer or does not, and the sample is large enough: $np_0 = 420{,}019(0.00034) = 142.8 \geq 15$ and $n(1-p_0) = 420{,}019(0.99966) = 419{,}876 \geq 15$. The first number is the "about 143" expected cases from the story.
2. **Hypotheses.** The parameter is p, the proportion of **all** cell phone users who develop brain cancer. The null value is the non-user rate, $p_0 = 0.034\% = 0.00034$. $H_0 : p = 0.00034$ (phones make no difference) and $H_A : p > 0.00034$ (users get brain cancer at a higher rate). The claim is about a **higher** rate, so the test is **upper-tailed**.
3. **Test statistic.** The sample proportion is $\hat{p} = 172/420{,}019 \approx 0.0004095$, or about 0.041%. The standard error uses p_0, because we work as if H_0 is true:

$$SE = \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.00034 \times 0.99966}{420{,}019}} \approx 0.00002845$$

$$z_{obs} = \frac{\hat{p} - p_0}{SE} = \frac{0.0004095 - 0.00034}{0.00002845} \approx 2.44$$

The users' cancer rate is 2.44 standard errors above the non-user rate.

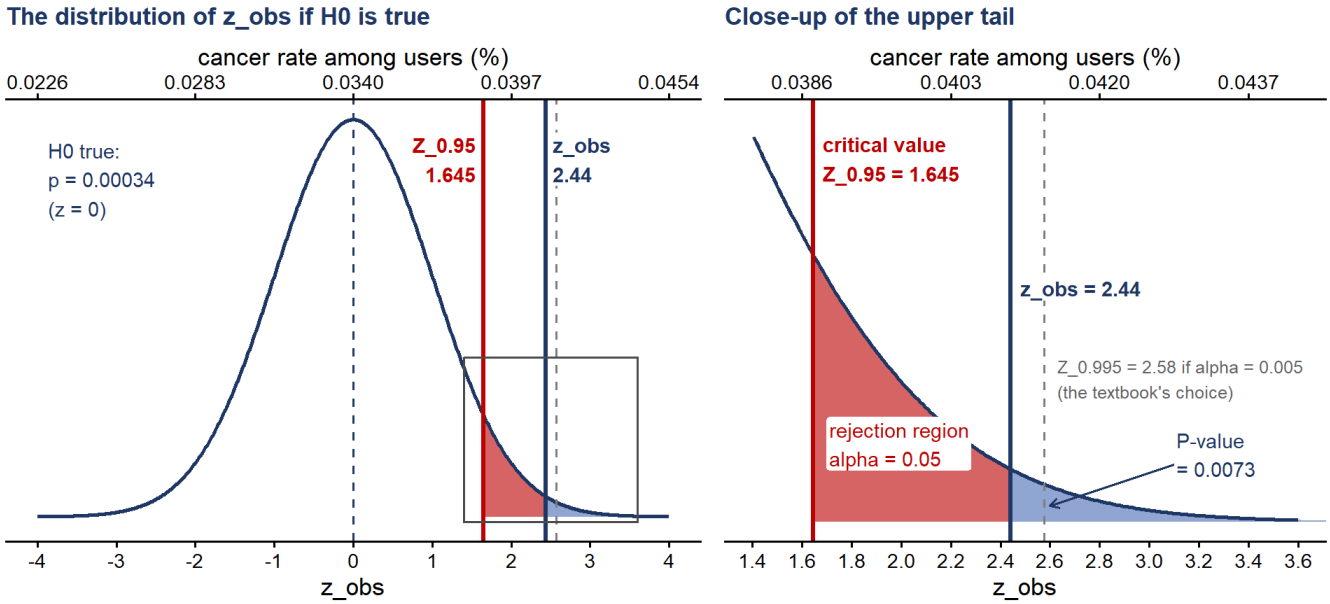
4. **P-value.** Upper-tailed, so we want the area to the right of z_{obs}:

$$P(Z \geq 2.44 \mid H_0 \text{ true}) = 1 - 0.9927 = 0.0073$$

In words: if phones made no difference, a study this size would see a cancer rate this far above 0.034% only about 73 times in 10,000.

5. **Decision.** $0.0073 < 0.05$: **reject** H_0. The data are convincing evidence that cell phone users develop brain cancer at a higher rate than non-users. (The study shows a higher rate among users. On its own it does not show that the phones are the **cause**.)

Here is the whole test as a picture. The curve is the distribution of z_{obs} **if** H_0 **is true** (phones make no difference). The **rejection region** is the upper 5% of that curve, everything at or beyond the **critical value** $Z_{1-\alpha} = Z_{0.95} = 1.645$, so $P(Z \geq 1.645 \mid H_0 \text{ true}) = 0.05 = \alpha$. (The test is upper-tailed, so all of α goes in one tail and the critical value is $Z_{1-\alpha}$. A two-sided test splits α between the tails and uses $Z_{1-\alpha/2}$.) Our test statistic, $z_{obs} = 2.44$, lands well inside it. The area beyond z_{obs} is the P-value, 0.0073, and it is smaller than α. That is the decision rule in picture form: z_{obs} lands in the rejection region exactly when the P-value is less than α.



The textbook's choice of α. OpenStax works this example at a much stricter $\alpha = 0.005$. Wrongly announcing that cell phones cause cancer would frighten a great many people and send research and regulation chasing a false cause, so the book asks for very strong evidence first. At that cutoff the critical value moves right to $Z_{0.995} = 2.58$ (the grey dashed line), $z_{obs} = 2.44$ falls just short of it, and $0.0073 \geq 0.005$: **do not reject H_0**. Same data, opposite decision. The decision depends on the cutoff, which is why α is chosen **before** the data are collected.

Assumptions for a population proportion

Intervals and tests for p both need a random sample, with each individual counted as a success or a failure. They differ in the sample-size check:

- **Confidence interval for p:** $n\hat{p} > 5$ and $n(1-\hat{p}) > 5$, checked with $\hat{p}$, because p is unknown.
- **Significance test for p:** $np_0 \geq 15$ and $n(1-p_0) \geq 15$, checked with p_0, because the test assumes H_0 is true.

Either way, the check is there to make the sampling distribution of $\hat{p}$ close to normal, so that the areas from the z-table are right.

Try it out

- A friend claims they can predict coin flips. In a test of 10 flips, they call **7** correctly. Write the hypotheses and find the P-value. Is the result significant at $\alpha = 0.05$? *Answer:* $H_A : p > 0.5$, $H_0 : p = 0.5$ (upper-tailed). Under H_0, $X \sim \text{Binom}(10, 0.5)$, so P-value $= P(X \geq 7 \mid H_0 \text{ true}) = (120 + 45 + 10 + 1)/1024 = 176/1024 \approx 0.172$. $0.172 \geq 0.05$: do not reject H_0. Seven of ten is quite common for a guesser.
- Write H_0 and H_A, give θ_0, and name the type of test.
    - A coffee shop says its large cup holds 16 oz on average. Customers suspect it holds less. *Answer:* $H_0 : \mu = 16$, $H_A : \mu < 16$, $\mu_0 = 16$ oz, lower-tailed.
    - A website's sign-up rate used to be 12%. After a redesign, has it changed? *Answer:* $H_0 : p = 0.12$, $H_A : p \neq 0.12$, $p_0 = 0.12$, two-sided.
- A test statistic comes out to $z_{obs} = -2.10$. Find the P-value if H_A is lower-tailed, and if it is two-sided. What is the decision for each at $\alpha = 0.05$? At $\alpha = 0.01$? *Answer:* lower-tailed: $P(Z \leq -2.10) = 0.0179$; two-sided: $2 \times 0.0179 = 0.0358$. Both are below 0.05,

so reject H_0 either way at $\alpha = 0.05$. Neither is below 0.01, so do not reject H_0 at $\alpha = 0.01$.

- *(The Lecture 23 practice problem.)* Last year 30% of students skipped breakfast. This fall, campus dining asks a random sample of 400 students, and 136 say they skip it. Has the proportion gone up? Use $\alpha = 0.05$. *Answer:* $np_0 = 120$ and $n(1 - p_0) = 280$, both at least 15. $H_0 : p = 0.30$, $H_A : p > 0.30$ (upper-tailed). $\hat{p} = 136/400 = 0.34$, $SE = \sqrt{0.30 \times 0.70/400} \approx 0.0229$, $z_{obs} = 0.04/0.0229 \approx 1.75$. P-value $= 1 - 0.9599 = 0.0401 < 0.05$: reject H_0. The proportion of students skipping breakfast has gone up.

Testing a population mean

When the variable is quantitative and the claim is about its mean, $H_0 : \mu = \mu_0$, the five steps become:

1. **Assumptions.** A random sample, from a population that is normal or close to it. With $n \geq 30$ the Central Limit Theorem makes $\bar{x}$ close to normal anyway.
2. **Hypotheses.** $H_0 : \mu = \mu_0$ against $H_A : \mu > \mu_0$, $\mu < \mu_0$ or $\mu \neq \mu_0$.
3. **Test statistic.** Just as for confidence intervals, which one depends on whether σ is known:

$$\sigma \text{ known: } z_{obs} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \qquad \sigma \text{ unknown: } t_{obs} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$$

4. **P-value.** The same tails as for p. Upper-tailed: $P(T \geq t_{obs} \mid H_0 \text{ true})$; lower-tailed: $P(T \leq t_{obs} \mid H_0 \text{ true})$; two-sided: double the tail beyond t_{obs}. With z_{obs}, read it from the z-table as before.
5. **Decision.** Compare the P-value with α, then say what it means in context.

P-values from the t-table. The course t-table (misc/t_table.html) lists critical values, not areas, so it gives the P-value only between two bounds. In row df, find the two values that $|t_{obs}|$ falls between. The P-value is between the upper-tail areas at the top of those two columns (for a two-sided test, use the two-tail areas). Software gives the exact value.

Example: Marco's throwing distance (σ known)

From OpenStax Introductory Statistics 2e, Section 9.5, Try It 9.14 (https://openstax.org/books/introductory-statistics-2e/pages/9-5-additional-information-and-full-hypothesis-test-examples).

> The mean throwing distance of a football for **Marco**, a high school quarterback, is **40 yards**, with a standard deviation of **two yards**. The team coach tells Marco to adjust his grip to get more distance. The coach records the distances for **20** throws. For the 20 throws, Marco's mean distance was **45 yards**. The coach thought the different grip helped Marco throw farther than 40 yards. Conduct a hypothesis test using a preset $\alpha = 0.05$. Assume the throw distances for footballs are normal.

Marco's long record gives the population standard deviation, $\sigma = 2$, so this is a z-test. (The book leaves this one as an exercise; the solution below is ours.)

Question: At $\alpha = 0.05$, does the new grip make Marco throw farther than 40 yards, on average?

1. **Assumptions.** We treat the 20 throws as a random sample of Marco's throws with the new grip. $n = 20$ is small, but the distances are normal, so the test works. $\sigma = 2$ is known, so we use z.
2. **Hypotheses.** μ = Marco's mean throwing distance with the new grip. $H_0 : \mu = 40$ and $H_A : \mu > 40$. The coach thinks he throws **farther**, so the test is **upper-tailed**.
3. **Test statistic.**

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{20}} \approx 0.4472$$

$$z_{obs} = \frac{\bar{x} - \mu_0}{SE} = \frac{45 - 40}{0.4472} \approx 11.18$$

   If H_0 is true, $z_{obs} \sim N(0, 1)$ (standard normal).
4. **P-value.** $P(Z \geq 11.18 \mid H_0 \text{ true})$ is far off the end of the z-table: **essentially 0**.
5. **Decision.** P-value $\approx 0 < 0.05$: **reject** H_0. The data are convincing evidence that the new grip makes Marco throw farther than 40 yards, on average.

   With the critical value. The test is upper-tailed with $\alpha = 0.05$, so the critical value is $Z_{1-\alpha} = Z_{0.95} = 1.645$. The rejection region is $z_{obs} \geq 1.645$. Since $z_{obs} = 11.18 \geq 1.645$, it lands in the rejection region: the same decision, reject H_0.

Example: the first statistics test (σ unknown)

From OpenStax Introductory Statistics 2e, Section 9.5, Example 9.16 (https://openstax.org/books/introductory-statistics-2e/pages/9-5-additional-information-and-full-hypothesis-test-examples).

> Statistics students believe that the mean score on the first statistics test is **65**. A statistics instructor thinks the mean score is **higher** than 65. He samples **ten** statistics students and obtains the scores
>
> 65, 65, 70, 67, 66, 63, 63, 68, 72, 71.
>
> He performs a hypothesis test using a 5% level of significance. The data are assumed to be from a normal distribution.

No population standard deviation is given, only the ten scores, so σ is unknown and this is a t-test.

Question: At $\alpha = 0.05$, is the mean score on the first statistics test higher than 65?

1. **Assumptions.** We treat the ten students as a random sample. $n = 10$ is small, but the scores are normal, so the test works. σ is unknown, so we use t.
2. **Hypotheses.** μ = the mean score of all statistics students on the first test. $H_0 : \mu = 65$ and $H_A : \mu > 65$. The instructor thinks the mean is **higher**, so the test is **upper-tailed**.
3. **Test statistic.** From the data, $\bar{x} = 670/10 = 67$ and $s = 3.197$.

$$SE = \frac{s}{\sqrt{n}} = \frac{3.197}{\sqrt{10}} \approx 1.011$$

$$t_{obs} = \frac{\bar{x} - \mu_0}{SE} = \frac{67 - 65}{1.011} \approx 1.98, \qquad df = 10 - 1 = 9$$

   If H_0 is true, $t_{obs} \sim t(9)$ (Student's t with 9 degrees of freedom).
4. **P-value.** Upper-tailed, so we want $P(T \geq 1.98 \mid H_0 \text{ true})$. In row 9 of the t-table (misc/t_table.html), $1.833 < 1.98 < 2.262$. Those columns have upper-tail areas 0.05 and 0.025, so

$$0.025 < P(T \geq 1.98 \mid H_0 \text{ true}) < 0.05$$

   Software gives 0.0395.
5. **Decision.** The P-value is below $\alpha = 0.05$: **reject** H_0. The data are convincing evidence that the mean score on the first test is higher than 65, as the instructor thinks.

With the critical value. The test is upper-tailed with $\alpha = 0.05$ and $df = 9$, so the critical value is $t_{9,\,0.95} = 1.833$ (row 9, upper-tail area 0.05). The rejection region is $t_{obs} \geq 1.833$. Since $t_{obs} = 1.98 \geq 1.833$, it lands in the rejection region: the same decision, reject H_0.

Example: the boiling point of a liquid (σ unknown)

From OpenStax Introductory Statistics 2e, Section 9.5, Try It 9.20 (https://openstax.org/books/introductory-statistics-2e/pages/9-5-additional-information-and-full-hypothesis-test-examples).

> The boiling point of a specific liquid is measured for **15** samples, and the boiling points are
>
> 205, 206, 206, 202, 199, 194, 197, 198, 198, 201, 201, 202, 207, 211, 205.
>
> Is there convincing evidence that the average boiling point is **greater than 200**? Use a significance level of 0.1. Assume the population is normal.

Again only the data are given, so σ is unknown and we use t. (The book leaves this one as an exercise; the solution below is ours.)

Question: At $\alpha = 0.10$, is the mean boiling point of this liquid greater than 200?

1. **Assumptions.** We treat the 15 measurements as a random sample. $n = 15$ is small, but the population is normal, so the test works. σ is unknown, so we use t.
2. **Hypotheses.** μ = the mean boiling point of the liquid. $H_0 : \mu = 200$ and $H_A : \mu > 200$: **upper-tailed**.
3. **Test statistic.** From the data, $\bar{x} = 3{,}032/15 \approx 202.13$ and $s \approx 4.533$.

$$SE = \frac{s}{\sqrt{n}} = \frac{4.533}{\sqrt{15}} \approx 1.170$$

$$t_{obs} = \frac{\bar{x} - \mu_0}{SE} = \frac{202.13 - 200}{1.170} \approx 1.82, \qquad df = 15 - 1 = 14$$

   If H_0 is true, $t_{obs} \sim t(14)$ (Student's t with 14 degrees of freedom).
4. **P-value.** In row 14 of the t-table (misc/t_table.html), $1.761 < 1.82 < 2.145$. Those columns have upper-tail areas 0.05 and 0.025, so

$$0.025 < P(T \geq 1.82 \mid H_0 \text{ true}) < 0.05$$

Software gives 0.045.

5. **Decision.** The P-value is below $\alpha = 0.10$: **reject** H_0. The data are convincing evidence that the mean boiling point of the liquid is greater than 200.

   With the critical value. The test is upper-tailed with $\alpha = 0.10$ and $df = 14$, so the critical value is $t_{14,\,0.90} = 1.345$ (row 14, upper-tail area 0.10). The rejection region is $t_{obs} \geq 1.345$. Since $t_{obs} = 1.82 \geq 1.345$, it lands in the rejection region: the same decision, reject H_0.

The next two problems are extra practice: one proportion and one mean.

Practice: households with three cell phones

From OpenStax Introductory Statistics 2e, Section 9.5, Example 9.18 (https://openstax.org/books/introductory-statistics-2e/pages/9-5-additional-information-and-full-hypothesis-test-examples).

> Suppose a consumer group suspects that the proportion of households that have **three cell phones** is **30%**. A cell phone company has reason to believe that the proportion is **not** 30%. Before they start a big advertising campaign, they conduct a hypothesis test. Their marketing people survey **150** households, with the result that **43** of the households have three cell phones.
>
> a. Calculate the sample proportion.
> b. What is a success for this problem?
> c. What is the level of significance?
> d. Draw the graph for this problem and calculate the P-value.
> e. Make a decision: reject or do not reject H_0, and why?

No significance level is given, so we use the usual $\alpha = 0.05$.

1. **Assumptions.** A random sample of households, and a large enough sample: $np_0 = 150(0.30) = 45 \geq 15$ and $n(1 - p_0) = 150(0.70) = 105 \geq 15$. A **success** is a household with three cell phones.
2. **Hypotheses.** p = the proportion of **all** households with three cell phones. $H_0 : p = 0.30$ and $H_A : p \neq 0.30$. The company thinks the proportion is "not 30%", in either direction, so the test is **two-sided**.

3. **Test statistic.** $\hat{p} = 43/150 \approx 0.2867$. Using p_0 in the standard error,

$$SE = \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.30 \times 0.70}{150}} \approx 0.0374$$

$$z_{obs} = \frac{\hat{p} - p_0}{SE} = \frac{0.2867 - 0.30}{0.0374} \approx -0.36$$

4. **P-value.** Two-sided, so we count both tails:

$$P(Z \leq -0.36 \text{ or } Z \geq 0.36 \mid H_0 \text{ true}) = 2 \times 0.3594 = 0.7188$$

   (The book's calculator, without rounding z_{obs}, gives 0.7216.) The graph is the standard normal curve with both tails beyond ± 0.36 shaded: most of the curve.
5. **Decision.** $0.7188 \geq 0.05$: **do not reject** H_0. The data are not convincing evidence that the proportion of households with three cell phones differs from 30%.

Practice: faster with goggles?

From OpenStax Introductory Statistics 2e, Section 9.5, Example 9.14 (https://openstax.org/books/introductory-statistics-2e/pages/9-5-additional-information-and-full-hypothesis-test-examples).

When Jeffrey was eight years old, he established a mean time of **16.43 seconds** for the 25-yard freestyle, with a standard deviation of $\sigma = 0.8$ seconds. His dad, Frank, thinks a good pair of goggles would make Jeffrey faster, so he buys him an expensive new pair.

Frank times Jeffrey on **15** swims of the 25-yard freestyle in the new goggles. The mean of those 15 times is **16 seconds**, almost half a second faster than usual. Did the goggles help, or did Jeffrey just have a good set of swims? Jeffrey's swim times are normally distributed, and Frank uses $\alpha = 0.05$.

Question: At $\alpha = 0.05$, do the data show that Jeffrey swims faster in the new goggles?

1. **Assumptions.** We treat the 15 swims as a random sample of Jeffrey's swims in the goggles. $n = 15$ is small, but the swim times are normal, so the test works. $\sigma = 0.8$ is known, so we use z, not t.
2. **Hypotheses.** μ = Jeffrey's mean 25-yard freestyle time in the new goggles. $H_0 : \mu = 16.43$ (the goggles make no difference) and $H_A : \mu < 16.43$. **Faster means less time**, so the test is **lower-tailed**.

3. **Test statistic.** $\bar{x} = 16$.

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{15}} \approx 0.2066$$

$$z_{obs} = \frac{\bar{x} - \mu_0}{SE} = \frac{16 - 16.43}{0.2066} \approx -2.08$$

   Jeffrey's mean in the goggles is about 2.08 standard errors below his usual mean.
4. **P-value.** Lower-tailed, so we want the area to the left of z_{obs}:

$$P(Z \leq -2.08 \mid H_0 \text{ true}) = 0.0188$$

   If the goggles made no difference, a set of 15 swims would average this fast only about 2 times in 100.
5. **Decision.** $0.0188 < 0.05$: **reject** H_0. The data are convincing evidence that Jeffrey swims the 25-yard freestyle faster in the new goggles.

Sources

- Coin flips: Bartoš, F., et al. (2025), "Fair coins tend to land on the same side they started: Evidence from 350,757 flips", *Journal of the American Statistical Association* 120(552), 2118–2127 (preprint on arXiv (https://arxiv.org/abs/2310.04153)). The physics model: Diaconis, P., Holmes, S., and Montgomery, R. (2007), "Dynamical bias in the coin toss", *SIAM Review* 49(2), 211–235.
- The taste test, Candidate X's poll, the chip bag, the coffee shop and the website are made-up examples.
- Cell phones and brain cancer: OpenStax *Introductory Statistics 2e*, Example 9.21. The first statistics test, the households with three cell phones and the goggles are Examples 9.16, 9.18 and 9.14; Marco is Try It 9.14 and the boiling point is Try It 9.20 (our solution). The breakfast survey is made up.
- Textbook: OpenStax *Introductory Statistics 2e*, Chapter 9, Sections 9.1–9.6.